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mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE Books (
  -> book_id INT PRIMARY KEY,
  -> title VARCHAR(100) NOT NULL,
  -> isbn VARCHAR(20) UNIQUE,
  -> published_year INT CHECK (published_year < 2027)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
  -> (1, 'The Great Gatsby', '9780743273565', 1925),
  -> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
  -> (3, '1984', '9780451524935', 1949);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title           | isbn           | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby | 9780743273565 | 1925          |
| 2       | To Kill a Mockingbird | 9780061120084 | 1960          |
| 3       | 1984            | 9780451524935 | 1949          |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> CREATE TABLE Members (
  -> member_id INT PRIMARY KEY,
  -> full_name VARCHAR(100),
  -> email VARCHAR(100) UNIQUE
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Members (member_id, full_name, email) VALUES
  -> (101, 'John Smith', 'john.smith@email.com'),
  -> (102, 'Emma Wilson', 'emma.wilson@email.com'),
  -> (103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name      | email              |
+-----+-----+-----+
| 101       | John Smith     | john.smith@email.com |
| 102       | Emma Wilson    | emma.wilson@email.com |
| 103       | Michael Brown  | michael.brown@email.com |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> Fig 4: Members Result Grid - arrow highlights the UNIQUE email column
-> Explanation:
  -> • PRIMARY KEY on member_id gives each member a unique identity (101, 102, 103).
  -> • UNIQUE on email enforces one account per email - a second signup with john.smith@email.com would
  -> fail with ERROR 1062.

mysql> CREATE TABLE Members (
  -> member_id INT PRIMARY KEY,
  -> full_name VARCHAR(100),
  -> email VARCHAR(100) UNIQUE
  -> );
ERROR 1050 (42S01): Table 'members' already exists
mysql> INSERT INTO Members (member_id, full_name, email) VALUES
  -> (101, 'John Smith', 'john.smith@email.com'),
  -> (102, 'Emma Wilson', 'emma.wilson@email.com'),
  -> (103, 'Michael Brown', 'michael.brown@email.com');
ERROR 1062 (23000): Duplicate entry '101' for key 'members.PRIMARY'
mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name      | email              |
+-----+-----+-----+
| 101       | John Smith     | john.smith@email.com |
| 102       | Emma Wilson    | emma.wilson@email.com |
| 103       | Michael Brown  | michael.brown@email.com |
+-----+-----+-----+
3 rows in set (0.00 sec)
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mysql> CREATE TABLE Loans (
-> loan_id INT PRIMARY KEY,
-> member_id INT,
-> book_id INT,
-> loan_date DATE,
-> FOREIGN KEY (member_id) REFERENCES Members(member_id),
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
-> (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
-> (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
-> (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
-> (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
-> (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
Query OK, 10 rows affected (0.05 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Loans;SELECT m.full_name AS Member_Name, b.title AS Book_Title
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 102 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
| 6 | 103 | 2 | 2025-02-12 |
| 7 | 101 | 3 | 2025-03-01 |
| 8 | 102 | 3 | 2025-03-07 |
| 9 | 103 | 1 | 2025-03-15 |
| 10 | 101 | 1 | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

-> FROM Loans l
-> INNER JOIN Members m ON l.member_id = m.member_id
-> INNER JOIN Books b ON l.book_id = b.book_id;
+-----+-----+
| Member_Name | Book_Title |
+-----+-----+
| John Smith | The Great Gatsby |
| John Smith | To Kill a Mockingbird |
| John Smith | 1984 |
| John Smith | The Great Gatsby |
| Emma Wilson | To Kill a Mockingbird |
| Emma Wilson | The Great Gatsby |
| Emma Wilson | 1984 |
| Michael Brown | 1984 |
| Michael Brown | To Kill a Mockingbird |
| Michael Brown | The Great Gatsby |
+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT published_year, COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
+-----+-----+

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3 rows in set (0.01 sec)

mysql> CREATE TABLE Donation_History (
-> donation_id INT PRIMARY KEY,
-> book_id INT,
-> donor_name VARCHAR(100),
-> donation_date DATE,
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
-> VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE INDEX idx_books_isbn ON Books(isbn);
Query OK, 0 rows affected (0.09 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Books WHERE isbn = '9780451524935';
mysql> SELECT * FROM Books WHERE isbn = '9780451524935';
+-----+-----+-----+-----+
| book_id | title | isbn          | published_year |
+-----+-----+-----+-----+
|      3 | 1984  | 9780451524935 |      1949      |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |

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